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MIT.NANO

MASSACHUSETTS INSTITUTE OF TECHNOLOGY -AN AIA COTE TOP 10 CASE STUDY

MIT.nano successfully turned MIT's nanotechnology, materials, and engineering systems into a single sustainable research center for health, energy, computing, and other emerging fields. This green building project focused on addressing humanity's greatest challenges, including sustainable energy, clean water, clean air, and even finding a cure for cancer! The project incorporated sustainable strategies to develop a design that is highly transparent, energy-efficient and inviting for the community and potential stakeholders. Overall, the sustainable design and regenerative land use make MIT.nano a model for future campus planning. Before we move ahead and discuss the various sustainable measures and standards used to review the green building project, let's credit everyone involved.

"Image Source: www.mit.edu"

Project title	**Massachusetts Institute of Technology | MIT.nano**
LEED Status	**LEED Platinum**
Awards/Credits	**2021 AIA COTE Top Ten Award** **2019 Lab of the Year Award** **AIA Education Facility Design Award of Merit** **ACEC Grand Award for Engineering Excellence** **I²SL Go Beyond Award** **Boston Society for Architecture Honor Award**
Architect	**HGA**
Owner	**Massachusetts Institute of Technology**
Location	**Cambridge, Massachusetts**
Project site	**Brownfield**
Building program type	**Education - College/University (campus-level), Laboratory**
Year of design completion	**2014**
Year of substantial project completion	**2018**
Gross conditioned floor area	**216,000 sq. ft.**
Number of stories	**7**
Project climate zone	**ASHRAE 5A**
Annual hours of operation	**8,760**
Site area	**101,110 sq. ft.**
Project site context/setting	**Urban**
Cost of construction, excluding furnishing	**$291,000,000**
Number of residents, occupants, visitors	**2,000**

- Architect: HGA
- Consultant (Audio-Visual): Cavanaugh Tocci Associates
- Consultant (Sustainability): BR+A Consulting Engineers, Inc.
- Consultant (Cleaning Planning/Tools): AGI Abbie Gregg, Inc.
- Consultant (Cleanroom): Cleanroom Construction Associates
- Consultant (Code): Jensen Hughes
- Consultant (Cost): Fennessy Consulting Services
- Consultant (Interiors): MoharDesign
- Consultant (Lighting): Ripman Lighting Consultants
- Consultant (Vibration/Acoustic): Acentech, Inc.
- Engineer (Civil): Kleinfelder
- Engineer (MEP/FP): BR+A Consulting Engineers, Inc.
- Engineer (Geotechnical/Foundations): Haley & Aldrich Consultant
- Engineer (Process Benches): Research Facilities Design (RFD)
- Engineer (Process Engineering/TGMS): Hallam-ICS
- Engineer (Structural): LeMessurier Consultants, Inc.
- Lab Casework Manufacturer: New England Laboratory Casework Co., Inc.
- Lab Fume Hoods Manufacturer: New England Laboratory Casework Co., Inc. (Mott)
- Landscape Architect: Pressley Associates
- Consultant (EMI): Vitatech
- Consultant (Curtainwall): Curtainwall Design Consulting
- Consultant (Exterior Envelope): Anis Building Enclosure Consultants
- Manager / General Contractor: Turner Construction Company

The project was judged rigorously by a jury of five. We have Erica Cochran Hameen (Associate AIA) from Carnegie Mellon University, Lynn Simon (FAIA) from Google, Marlon Blackwell (FAIA) from Marlon Blackwell Architects, Michelle Amt (AIA) from VMDO Architects, and Renee Cheng (FAIA) from the University of Washington.

TEN STANDARDS OF AIA COTE TOP TEN AWARDS PROGRAM

The AIA COTE (Committee on the Environment) Top Ten Awards are given each year to the ten most innovative green building projects that set the standards in design and sustainability. It is one of the industry's most prestigious awards programs that reward sustainability and sustainable design excellence. The committee evaluates the green building projects using ten standards/measures which it has developed over time.

1.Integration (to ensure a thoughtful process that delivers both beauty and function)

2.Equitable communities (to ensure the solutions positively impact future occupants and the larger community)

3.Ecosystems (to ensure the solution mutually benefits human and nonhuman inhabitants)

4.Water (to ensure proper improvement in the quality and conservation of water)

5.Economy (to ensure the design adds value for the owners, occupants, community, and the planet)

6.Energy (to ensure the design eliminates dependence on fossil fuels without compromising the project's function)

7.Well-being (to ensure the design supports health and well-being for all people)

8.Resources (to ensure the design counts on sustainable and informed material and resource selection)

9.Change (to ensure the solution addresses the future risk and vulnerabilities)

10.Discovery (to ensure the project has lessons to learn for future designers, users, and operators)

Now that we are well-acquainted with the frameworks and standards of the AIA COTE Top Ten Awards program, let's use them to evaluate the performance of the 'MIT.nano' project.

Evaluating green building project performance using AIA COTE Measures

1.Integration

Located nearby the iconic Infinite Corridor and the Great Dome, MIT.nano is home to class 100 and 1,000 cleanrooms, imaging suites, and world-class laboratories. The project team aimed to achieve best-in-class performance in designing for maximum safety and occupant well-being, providing high flexibility and resilience, and making the site highly energy efficient.

The integrated sustainability approach to achieve the goals mentioned above helped the team revitalize the heart of the campus while maximizing collaboration and engagement across disciplines. The team reused an existing brownfield building site and transformed the service yard into an inspiring landscape. The project successfully developed a high-performance cleanroom design and achieved unprecedented overall sustainability.

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"Image Source: www.aia.org"

2.Equitable Communities

The project team identified students, faculty, leadership, campus sustainability and resiliency committee, environmental health and safety, central engineering, and city planning groups as the project stakeholders. They were engaged through numerous workshops and town-hall meetings and were encouraged to participate in the building program, sustainability goals, and physical planning.

MIT.nano has adequate gathering spaces like 'The Moment' and the 'East Lobby' to invite occupants and engage in contemplation. Such strategically built gathering spaces make it easier to engage with colleagues. The building has open fire stairs and clean stairs that encourage occupants to walk and engage in physical activity. Most importantly, MIT.nano is connected to other buildings on every level, which helps maximize connectivity across the campus.

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The presence of the 'Northwest Passage' art installation by Olafur Eliasson benefits people that are not directly associated with the project. It is a part of MIT's Percent of the Art program. Moreover, the North Corridor and Improbability Walk transformed the area into an inviting pedestrian and bicycle-friendly neighborhood.

"Image Source: www.mit.edu"

MANDATORY METRIC	SCORE
Walk	92
Transit	67
Bike	99
Number of parking spaces	0
Number of parking spaces required by local zoning code	0
Bike storage	Yes

ENCOURAGED METRIC	SCORE
Percentage of building occupants who commute via alternative transportation	86 percent
Average number of showers per occupant	0.6
Community access to the project	Partial access to the interior
On-site food production	No

"Image Source: www.aia.org"

MIT.NANO FORMS A NEW CAMPUS COURTYARD NAMED 'IMPROBABILITY WALK' ALONG THE MAIN GROUP BUILDINGS AND ANCHORED BY THE GREAT DOME. NAMED IN HONOR OF ONE OF MIT'S MOST INSPIRATIONAL FACULTY MEMBERS: THE LATE INSTITUTE PROFESSOR MILDRED "MILLIE" DRESSELHAUS.

3.Ecosystems

The project team had a massive challenge to overcome. The paved industrial brownfield site included a hazardous materials storage building that contaminated the soil. Transforming the site to retain, treat, infiltrate stormwater and regenerate the natural ecosystem was vital to MIT.nano's well-being. At the same time, the team had to meet MIT's urban heat-island reduction goals without tampering with the site's historical significance.

MIT's Stormwater and Landscape Ecology Masterplan effectively addresses issues like stormwater management, resilience planning, heat-island reduction, and more. MIT.nano's planning and design successfully fit within the master plan. The project team planted over 100 trees that added a projected 25,000 sq. ft. of vegetation.

The rainwater was collected, cleaned, and recharged through hardscape and vegetated areas - ensuring the water is well treated before it makes its way to the Charles River. The team began with only 7.9 percent of the site area supporting vegetation. Post project completion, it successfully vegetated over 35 percent of the site! Intentional design strategies like dark skies, bird safety, and specific regional environmental concerns were also addressed.

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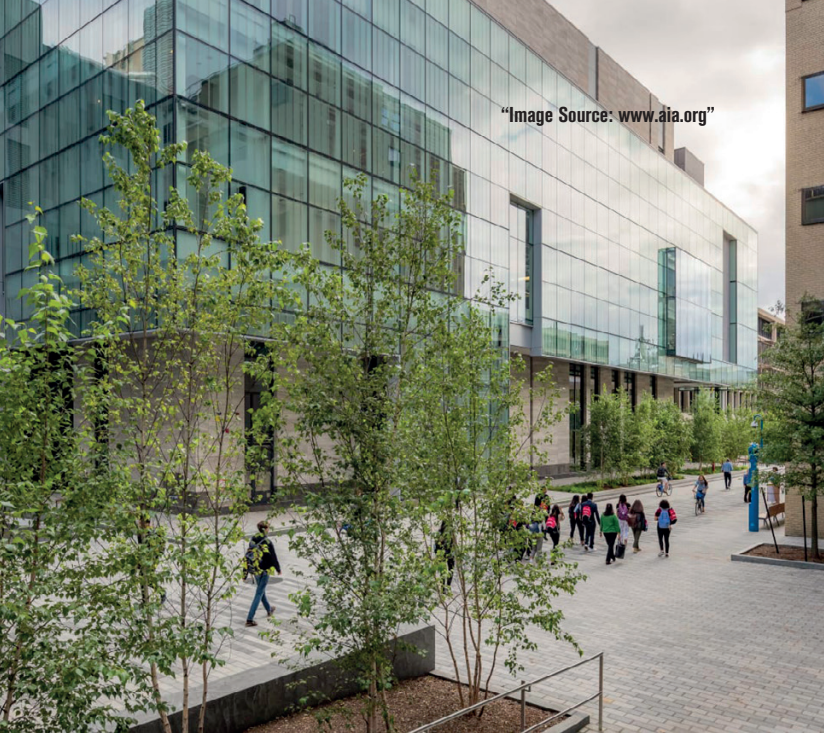
"Image Source: www.erieap.com"



THE BUILDING AND LANDSCAPE ARE DESIGNED FOR WATER CONSERVATION, DECREASING RUNOFF AND MITIGATION FOR FLOODING. STORMWATER IS MANAGED THROUGH STORAGE AND TREATMENT BELOW THE SURFACE.

4.Water

Since MIT's campus is surrounded by the Charles River and is near the Atlantic coast, the project site is built on tidal land reclaimed from the Charles River estuary. The area is covered in impervious surfaces and retains some of the river's drainage patterns. Therefore, the stormwater runoff flows through the drainage into the Charles River.

The project team has installed low-flow and low-flush fixtures to ensure that the building consumes at least 49 percent less potable water. Since almost 64 percent of the site area is covered in impervious surfaces, the site irrigation water depends entirely on the reverse osmosis reject reclaim system. Furthermore, MIT's new cooling towers (Central Utility Plant) significantly reduce particulate emissions and boost water conservation.

5.Economy

MIT.nano was also driven by strong economic factors. Multi-million-dollar investments made the project's core, which facilitates expensive high-performance space to monitor and regulate vibration, EMI, acoustics, temperature, and humidity. Most importantly, the team managed to establish such a facility at the heart of campus - eliminating the need for more of such investments.

SERVING 2000 RESEARCHERS ON CAMPUS AND THE COMMUNITY. BY INVESTING IN A SHARED CORE FACILITY MIT ELIMINATES DUPLICATION OF EXPENSIVE TOOLS ACROSS THE CAMPUS WHILE MAXIMIZING COLLABORATION.

The team used an intensive process to establish a life cycle cost analysis that included 140 Energy Conservation Measures (ECM). MIT.nano included sixty of the ECMs based on the highest ROI and largest carbon reduction impact. The project team also had to consider programmatic changes over a period of 50-100 years. Therefore, they built robust MEP systems that included 12 percent additional capacity beyond full-built need.

The selection of stone cladding and terrazzo floors helped reduce maintenance costs and keep the project profitable throughout its life cycle. The construction costs per square foot amounted to $1,271/sq. ft. Meanwhile, the team saved $5.2M in construction costs and successfully reduced embodied carbon.

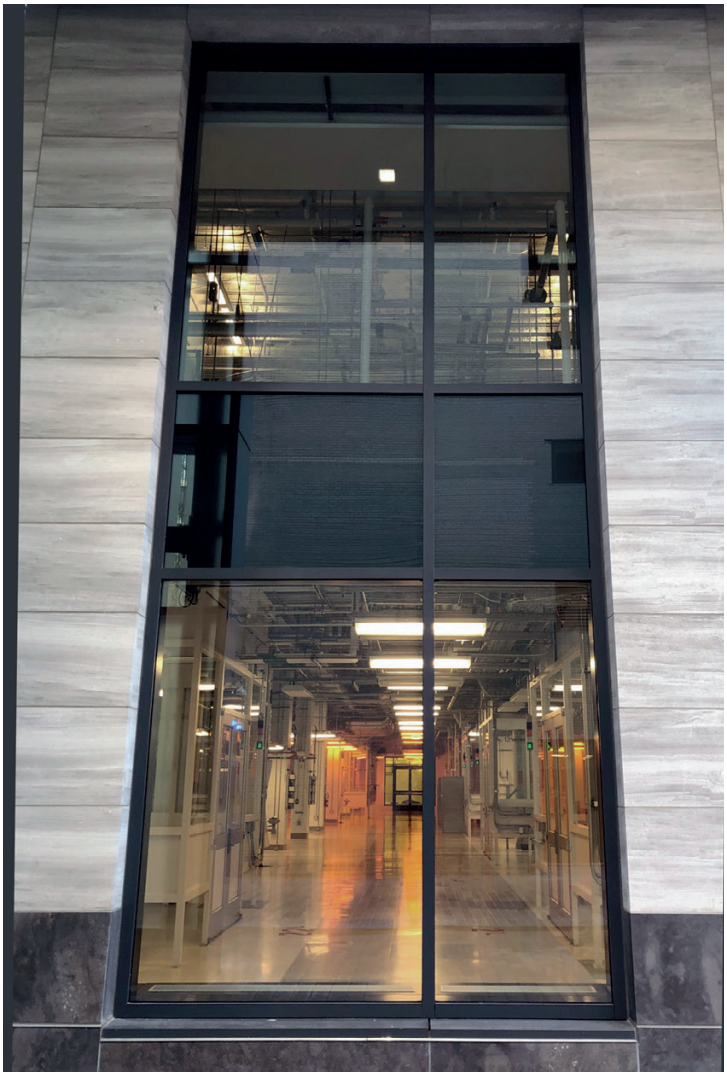
6.Energy

Research facilities and nanoscale laboratories are known for high energy consumption. To get the project LEED-certified, the team had to come up with innovative energy-efficient ideas. Moreover, MIT.nano required a particle-free 'cleanroom' environment that ticked all the boxes related to air filtration, temperature stability, humidity control, and ventilation.

The goal was simple - design the most energy-efficient academic cleanroom in the country. The team benchmarked several high-performance academic cleanrooms in the country and identified tool load assumptions, makeup-air (MAHU) efficiency, and recirculating-air (RAHU) efficiency as the three primary drivers of energy use.

The team used high-impact strategies like exhaust heat recovery, variable volume exhaust and makeup, 100 percent filter coverage of cleanroom ceiling, variable volume RAHU (based on demand), class 100 cleanroom air change rate below 165 ACH, and local reheat with low-temperature makeup air. As a result, MIT.nano achieved a 51 percent energy cost savings and a 50 percent reduction in greenhouse gas emissions compared to the LEED ASHRAE 90.1 - 2007 baseline.

MANDATORY METRICS	SCORE
Predicted gross EUI	753 kBtu/sf/yr
Measured gross EUI	184 kBtu/sf/yr
Zero Tool-Percent energy reduction	33 percent
Measured LPD	0.96 W/sf
Percent LPD reduction from IECC 2006	30 percent
Percent window-wall ratio	53 percent
Operational emissions	0.02 mT CO_2/sf/yr



"Image Source: www.erieap.com"

7. Well-being

Thermal, visual, aural, physical, and psychological comfort is the basis of occupant well-being. MIT.nano ensures safety for all occupants, especially within the hazardous environment of a cleanroom. The building has sensors to monitor temperature, airspeed, air filtration, and humidity.

It also has a high-performance curtainwall system and perimeter heating to reduce occupant discomfort. The project team also installed gas monitoring systems and ensured increased visibility to bring daylight deep into the lab. Glare and reflectivity were also significantly reduced across the building. The team used building materials that were specified to minimize Volatile Organic Compounds (VOCs) and risks to human health. The use of low-emitting materials helped the project earn a LEED credit. The team used a low-pressure drop design, sound attenuators, and vibration isolation to address the problem of airflow noise in the cleanroom.

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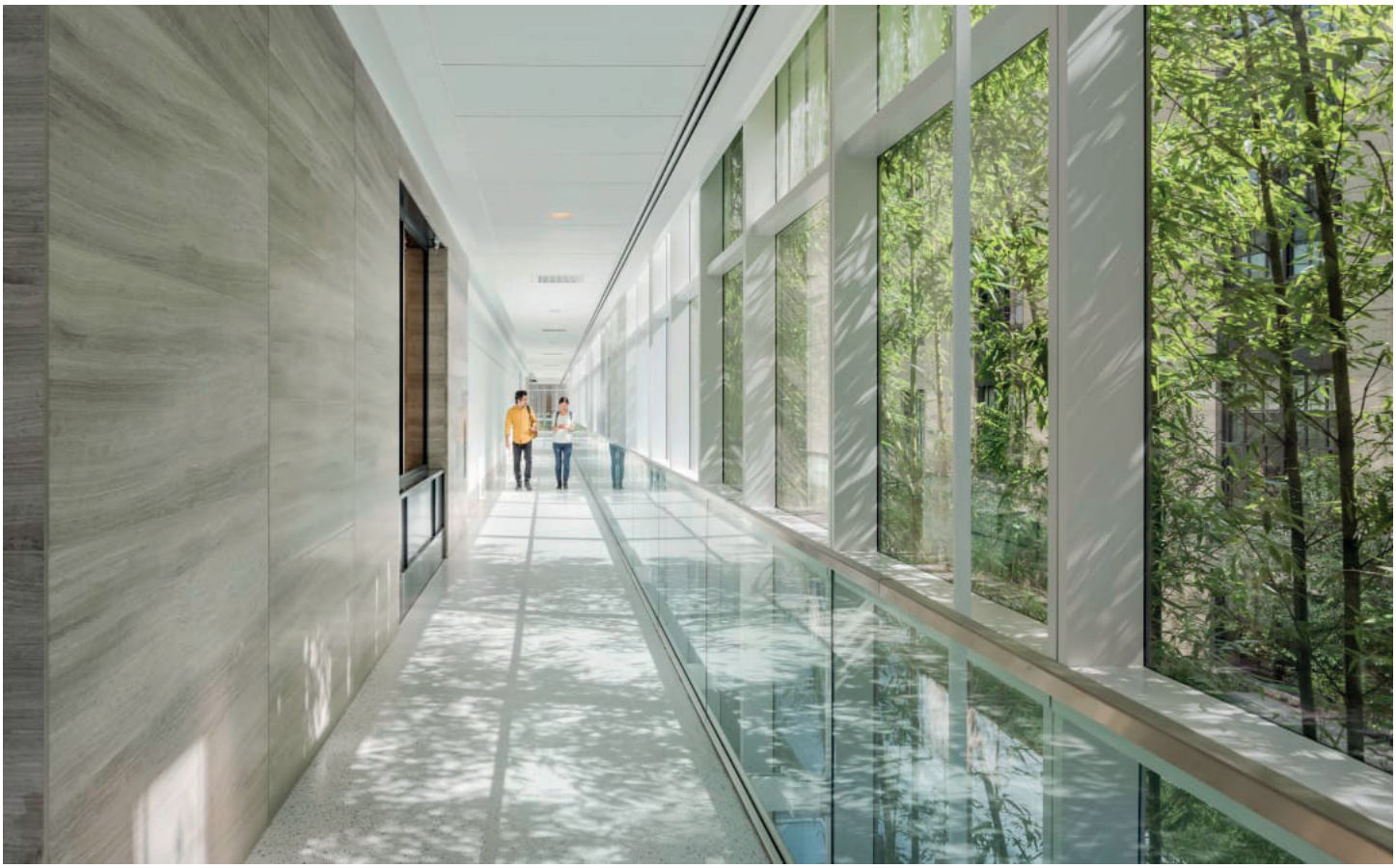
8.Resources

The selection of resources for MIT.nano focuses entirely on high durability and longevity. Although most nanoscale buildings are made with a cast-in-place reinforced concrete waffle slab structure, MIT.nano shifted to a steel frame primary structure with composite concrete metal floors and roof decks. This move helped the team reduce the materials tonnage and save $5.2M in construction costs.

MIT.nano has stone cladding for exterior and interior surfaces, fiber cement boards, and terrazzo floors. It also uses 100 percent FSC-certified lumber that lowers embodied carbon insulation. Additionally, the project used a construction waste management plan which resulted in the diversion of 90 percent of construction waste from landfills. This helped the project earn a LEED exceptional performance credit.



"Image Source: www.aia.org"

STONE WALL CLADDING AND TERRAZZO FLOORS ARE KEY MATERIALS USED IN PUBLIC AREAS. DURABILITY AND LONGEVITY WERE THE IMPORTANT MATERIALS SELECTION CRITERIA.

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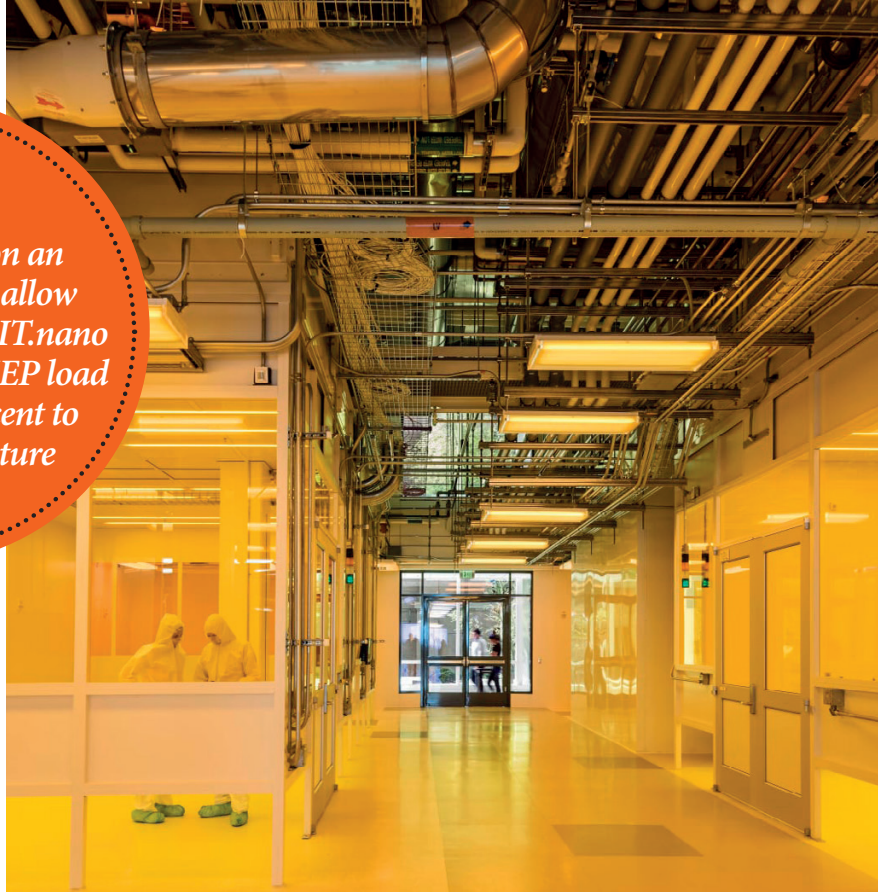
9.Change

MIT.nano set a benchmark when it came to designing for adaptability. Since it is a science research facility, it was of prime importance to plan considering the rapid changes in research, instrumentation, and other crucial factors. Therefore, the project team ensured the programmatic addition of a nano-makerspace for future research.

The team focused on a modular design on an 11-foot module to allow future variations. MIT.nano has an additional MEP load capacity of 12 percent to accommodate future loads.

The team focused on a modular design on an 11-foot module to allow future variations. MIT.nano has an additional MEP load capacity of 12 percent to accommodate future loads. The building is prepared for H5 (high hazard) occupancy, which allows it to set up nano-research labs on upper levels.

Furthermore, the team included floor mitigation measures like slurry wall construction, elevating ground-floor level by six inches, relocating electrical switchgear on upper levels, elevating the generator fuel oil pumps six feet above the basement slab and relocating the emergency generator in the penthouse.



THE BUILDING ANTICIPATES CHANGES IN THE SCIENCE AND RESEARCH FOCUS OVER TIME; THROUGH MODULAR DESIGN, GENEROUS FLOOR TO FLOOR HEIGHTS AND ROBUST MEP INFRASTRUCTURE. THE RESEARCH SPACE CAN BE READILY ADAPTED TO CHANGING NEEDS.

"Image Source: www.aia.org"

10.Discovery

The findings from this LEED Platinum project have been presented firm-wide. The case study has been shared with several participating institutions and professional conferences like I2SL, ESTECH, SCUP, and Tradeline. The project team is preparing to put the unique energy conservation measures implemented at MIT.nano for future projects at Purdue University and Northeastern University. Post-Occupancy Evaluation (POE) is being performed while the project continually disseminates information.

FINAL REMARKS

MIT.nano proves to be a benchmark for achieving sustainability goals while simultaneously maintaining MIT's state-of-the-art practices for research and cleanroom design. The project's commitment to advancing environmentally responsible design and planning helped turn the most energy-intensive building on the campus into the most energy-efficient. Besides the 2021 AIA COTE Top Ten award, MIT.nano was recognized for its significant achievements in advancing climate action and was awarded the 2019 Lab of the Year Award by R&D magazine, AIA Education Facility Design Award of Merit, ACEC Grand Award for Engineering Excellence, I²SL Go Beyond Award by the International Institute for Sustainable Laboratories, and Boston Society for Architecture Honor Award.

SOURCES:

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